Loss Characterization of Waveguides in Lithium Niobate on Insulator

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***Abstract*— Single crystal lithium niobate is a commonly used in optical electronic applications due to its variety of optical effects. However, it is a fairly difficult material to pattern and etching typically induces rough sidewalls. As a result of these limitations, commercial devices are typically use diffusion-type processes which only creates a small index increase. These devices have large footprints and therefore are costly. Lithium niobate on insulator (LNOI) is a technology created to circumvent some of these problems. It consist of a thin film of lithium niobate bonded to an insulating layer, which is then bonded to a host substrate. This allows compact structures due to the higher index confinement achievable. However, propagation loss is still prohibitive, due to the roughness induced by etching processes. We seek to characterize the propagation loss of waveguides in LNOI, and also to reduce the propagation losses.**

Keywords— Lithium Niobate, Optical Waveguides, Photonics

I. Introduction

Single crystal lithium niobate ($LiNbO_3$) is a very useful optical material as it have very strong optical effects. [1] This includes the very useful Pockels effect, Kerr effect and sum frequency generation. Utilization of these effects is critical for both current commercial devices and future quantum optics applications. The fundamental building block of optical devices is the waveguide, and to create waveguides in $LiNbO_3$ refractive index guiding is typically used. However, the two most common fabrication techniques to increase the refractive index in $LiNbO_3$, titanium diffusion and proton exchange, have a very small associated refractive index increase. This prevents compact structures due to prohibitive bending losses. As a result, photonic devices in lithium niobate exhibit large footprints, and this exacerbates the cost. Following the idea of 'smart-cut silicon', Lithium Niobate on Insulator (LNOI)[2] is a wafer processing technology developed, which provides a thin film of single crystal lithium niobate on top of a host substrate (typically SiO_2 grown on top of a bulk lithium niobate crystal). The advantages are similar to silicon-on-insulator (SOI), in that optical confinement in the vertical direction is provided by the high refractive index contrast between $LiNbO_3$ and SiO_2. By etching the thin film of lithium niobate, we can then also obtain high index contrast in the horizontal direction. As a result of the high index contrast attainable, compact optical structures can be fabricated with lower loss than before. However, due to the etching-induced rough sidewalls, propagation loss are still typically quite high. This can be seen in microscope images of waveguides with light coupled into it; the sidewalls are much brighter than the inside of the waveguides. In this paper, we seek to characterize loss in LNOI waveguides, and also explore methods to reduce propagation loss. [3]

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II. Device Design

We will utilize two main techniques to measure the loss. The first technique utilizes the Fabry Perot resonances.[4] The edges of the lithium niobate crystal form a weak cavity, and this will cause resonances to occur within the crystal, that corresponds to the length of the sample. The ratio between the high and low power can provide information about the loss, and the period between peaks can provide information about the effective index. Unfortunately, the Fabry Perot method is dependent on good estimation of the facet reflectivity, which in turn is dependent upon the edge quality, and may incorrectly estimate the propagation loss. Therefore, as our second method we also use an optical method to measure the loss. Using Y-branch splitters, we split each input into 4 separate waveguides (as shown in Figure 1), then place gratings to couple light out at several different known lengths. The light is imaged onto a CCD camera, and the power output at each grating can therefore be estimated. The differences in light output at each grating is therefore dependent on the additional length of waveguide traveled, thus the ratios can provide us information about the loss.

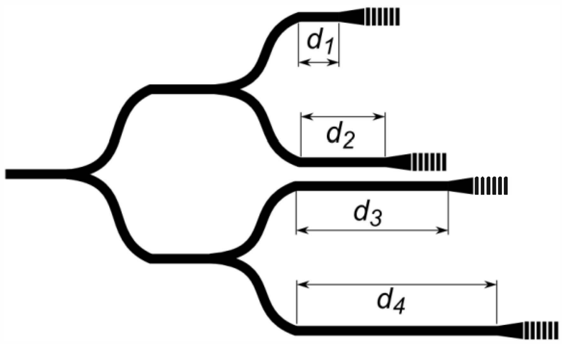

Fig. 1. Gratings at periodic intervals to measure the loss, with known distances d_1 to d_4

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We designed the gratings by parameter sweeping the period and duty cycle in a finite-difference time-domain software. We then collect the flux from an area directly above the gratings. From the simulations, an appropriate duty and period is chosen to maximize the light coupling straight upwards. From the simulation results (Figure 2), small periods and low duty ratio seemed to be preferred for light going upward. However small feature are intolerant to fabrication inaccuracies, so we prefer larger period and more balanced duty ratio. We choose 1100 nm and 35% duty as a compromise between fabrication capabilities and light coupled out. In fact, with larger features, the device is more tolerant to error, and additionally since the same fabrication process applies to all the gratings, we can expect the ratio between each grating to accurately represent the propagation loss.

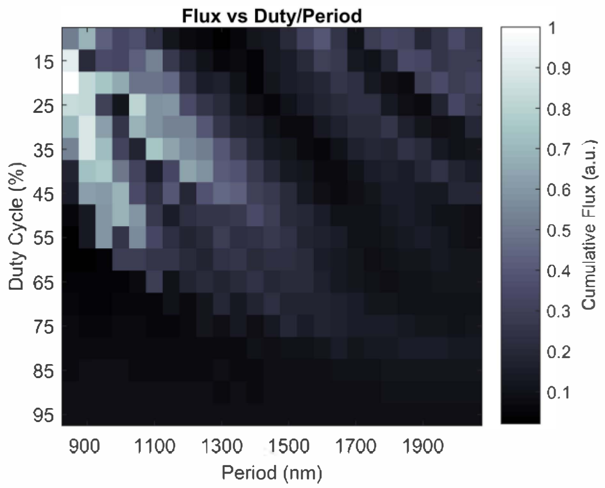

Fig. 2. Cumulative flux measured at a region above the grating.

III. Fabrication Methodology

Electron beam lithography (EBL) is used to fabricate the mask. To create stitch-free long waveguides, we use Fixed Beam Moving Stage (FBMS) by Raith, wherein the beam generates a circular pattern, and the stage then scans the intended device.[5] This allows us to create stitch-free devices spanning the entire chip. We deposit a layer of chromium (60nm) over the chip before spin-coating the resist. After development, we etch the chip in chromium etchant. This creates bi-layer mask of resist and chromium, allowing us to etch more deeply, as the selectivity of our etching methods are not particularly high. The devices are etched using argon ion milling, and this purely physical process creates a higher smoothness than reactive ion based processes. Both the patterning and milling were optimized to reduce sidewall roughness, particularly taking suggestions from [6]. Subsequently, the masks are removed and then the chip is diced to ensure an optically smooth end facet. To illustrate the working Y-branch 50/50 splitter, we coupled a supercontinuum source into it (shown in Figure 3).

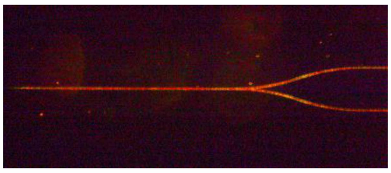

Fig. 3. Y-Branch splitter illuminated by supercontinuum laser, demonstrating 50/50 power split.

IV. Results

Using a tunable laser, we measured the transmission spectrum. We then used a program to calculate automatically the loss via the Fabry Perot effect. We used a mode solver to calculate the TM effective index (1.896) to calculate the Fresnel reflection at the facet. There are some variation throughout the spectrum, but the loss tends to be average around 3 dB/cm for a 3 µm waveguide. This is shown in Figure 4 below.

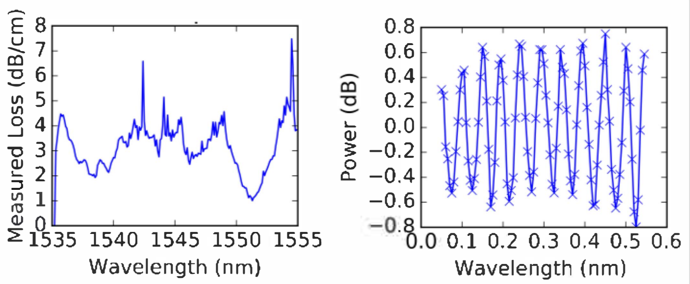

Fig. 4. Output Fabry-Perot resonance. LHS: loss calculated for each point, by calculating locally the maximum and minimum power for the Fabry Perot resonance effect. RHS: transmission spectrum around 1551 nm (starting from 1550.7nm).

References

[1] T. A. Ramadan, M. Levy, and R. M. Osgood, "Electro-optic modulation in crystal-ion-sliced z-cut LiNbO3 thin films," *Applied Physics Letters,* vol. 76, pp. 1407-1409, 2000.

[2] A. Guarino, G. Poberaj, D. Rezzonico, R. Degl'Innocenti, and P. Gunter, "Electro-optically tunable microring resonators in lithium niobate," *Nat Photon,* vol. 1, pp. 407-410, 2007.

[3] S. Y. Siew, S. S. Saha, M. Tsang, and A. J. Danner, "Rib Microring Resonators in Lithium Niobate on Insulator," *IEEE Photonics Technology Letters,* vol. 28, pp. 573-576, 2016.

[4] G. Tittelbach, B. Richter, and W. Karthe, "Comparison of three transmission methods for integrated optical waveguide propagation loss measurement," *Pure and Applied Optics: Journal of the European Optical Society Part A,* vol. 2, p. 683, 1993.

[5] J. E. Sanabia, K. E. Burcham, J. Klingfus, G. Piaszenski, M. Kahl, and R. Jede, "Fixed Beam Moving Stage electron beam lithography of waveguide coupling device structures," in *Lasers and Electro-Optics (CLEO), 2012 Conference on*, 2012.

[6] G. Ulliac, V. Calero, A. Ndao, F. I. Baida, and M. P. Bernal, "Argon plasma inductively coupled plasma reactive ion etching study for smooth sidewall thin film lithium niobate waveguide application," *Optical Materials,* vol. 53, pp. 1-5, 3// 2016.

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